

30	Which nuclide has the greatest initial activity?			
		nuclide	amount/mole	half-life/day
	A	$^{228}_{88}\text{Ra}$	0.7	2100
	B	$^{225}_{89}\text{Ac}$	0.002	10
	C	$^{228}_{90}\text{Th}$	0.09	400
	D	$^{241}_{94}\text{Pu}$	0.09	4800

Answer: A

$$A = \lambda N = (\ln 2 / t_{1/2}) N, \text{ where } N = n.N_A \text{ (i.e. } N \propto n)$$

$$A \propto \frac{N}{t_{1/2}} \propto \frac{n}{t_{1/2}}$$

Of the four options, option A gives the largest ratio $\frac{n}{t_{1/2}}$ hence greatest activity.

Option A: 3.33×10^{-4}

Option B: 0.0002

Option C: 0.000225

Option D: 0.00001875